Smaller explicit Diophantine forms for Con(ZFC)

Ten integer unknowns, compact quadratic systems, and a smaller fixed coefficient instance

Research synthesis prepared for the ProveIt repository

October 2026

Source snapshot `8c6eec9d19ef80538c8e61ff6e0180040583446a`

Abstract

We give explicit polynomial equations whose absence of solutions is provably equivalent in ZFC to its syntactic consistency. The fixed instance is derived from ProveIt's calibrated blank-input K215 machine; no unspecified recursively enumerable index or existential coefficient is used. A specialization of Sun's ten-integer construction, together with a coprime square merger and a quintic divisibility merger, gives an equation in ten unrestricted integer variables. A separately developed Jones-based equation uses sixteen unrestricted integer variables and 1,617 binary arithmetic operations. For readers prioritizing degree or evaluation cost, we give a system of 22 quadratic equations in 31 positive variables evaluated in 60 operations, its 125-operation quartic sum of squares, and a 12-positive-variable system evaluated in 102 operations. All these compatible forms can use a new complete native coefficient tuple whose largest port has 1,070,004 bits, 45.36% below the compared repository instance. The archive contains ordinary mathematical proofs, literal coefficient data, division-free circuits and reproducible finite audits. It does not claim global optimality or a new end-to-end proof-assistant certificate.

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1 The target and the meaning of "simpler"

Let Con(ZFC) mean that there is no finite derivation of a contradiction from the recursively presented first-order ZFC axioms, in the proof calculus used by the repository's K215 calibration. Equivalent standard effective proof calculi give provably equivalent consistency statements. The required conclusion has the following form for one fully fixed $F \in \mathbb{Z}[X_1, \ldots, X_n]$:

$$\mathrm{ZFC} \vdash \quad [\neg\exists \boldsymbol{x} \in \mathbb{Z}^n \; F(\boldsymbol{x}) = 0] \iff \mathrm{Con(ZFC)}. \tag{1.1}$$

This is a proof of a biconditional. It does not prove that either side holds. Under the usual consistency hypothesis, Gödel's second incompleteness theorem precludes ZFC from proving the left side once the biconditional has been established.

There are several inequivalent notions of simplicity. Fewer variables can cost enormous degree; a low-degree system can require more witnesses; small arithmetic circuits can contain very large fixed integers. A positive-variable equation also differs from an equation in unrestricted integers. We therefore retain several useful points instead of calling one construction an absolute minimum.

1.1 Concrete improvements over the pinned collection

All comparisons are against the immutable snapshot in [1], not against a moving branch. "Operation" means a binary addition, subtraction or multiplication in the emitted directed acyclic graph. Sharing is allowed; all powers of variables are expanded into such gates. Fixed integers are leaves. A system is evaluated as shared left and right sides to be compared; the comparisons themselves are not arithmetic gates. A sum of squared residuals includes the subtraction, squaring and summation gates. These conventions match the compared repository receipts.

Form	Domain	Variables	Degree	Operations
New Sun/Jones synthesis	$\mathbb{Z}$	10	d_{10}	1,442
Jones wrapper, package 29	$\mathbb{Z}$	17	d_{17}	20,607
New Jones wrapper	$\mathbb{Z}$	16	d_{16}	1,617
Quadratic system, package 30	$\mathbb{Z}_{>0}$	31	2	65
New quadratic system	$\mathbb{Z}_{>0}$	31	2	60
Quartic, package 30	$\mathbb{Z}_{>0}$	31	4	130
New quartic	$\mathbb{Z}_{>0}$	31	4	125
Product, package 30	$\mathbb{Z}_{>0}$	12	1,081	111
New product evaluator	$\mathbb{Z}_{>0}$	12	1,081	108
System, package 30	$\mathbb{Z}_{>0}$	12	1,072	107
New shifted system	$\mathbb{Z}_{>0}$	12	1,072	102

Here

$$d_{10} = 126496\,5^{31} + 27064,$$

$$d_{17} = 424296\,5^{33} + 53824, \qquad d_{16} = 813056\,5^{32} + 10016.$$

The new ten-integer equation is specified in full in Section 4. Its exact degree is

$$589042901992797851562527064,$$

and its evaluator uses 851 multiplications and 591 additions/subtractions. It combines the fixed quartic input with Sun's ten-integer reduction [10, 11] and the local simplifications proved there. The contribution is an explicit Con(ZFC) instance and smaller local formulas, not a new general ten-unknowns theorem.

The collection already contains nine-variable equations over $\mathbb{N}$ or $\mathbb{Z}_{>0}$. Ten unrestricted integer variables should not be compared with these as if the domains were identical. The existing 67-operation single equation and the 59-operation higher-degree system also remain better at their respective operation-count objectives. Our 60-operation system improves the quadratic/quartic point.

1.2 Fixed coefficients and the proof chain

All the new forms are tied to one complete native coefficient tuple. For its four ports the largest bit length falls from 1,958,376 to 1,070,004, while their combined raw storage falls from 977,220 to 533,099 bytes. These are measurements of the *native ports*; they are not a claim about the largest coefficient after expanding an entire high-degree polynomial. In particular, the sixteen-variable Jones wrapper has a much larger fixed base coefficient specified by a bounded power-of-two recipe.

The logical composition is

$$\begin{aligned} \text{a zero of the displayed equation/system} &\longleftrightarrow \text{a positive zero of the native parent} \\ &\longleftrightarrow \text{a finite blank-input K215 halt} \\ &\longleftrightarrow \text{a finite ZFC refutation.} \end{aligned}$$

Negating this equivalence yields (1.1). The parent-to-machine direction uses the native compiler and arithmetic-skeleton proofs in the pinned repository. The last equivalence is calibrated by `blank_halts_iff_zfc_refutation` and `nonhalting_iff_zfc_consistency` in `KEndpoint215.lean`. The replacement tuple is justified by the ordinary adapter proof in Section 2.

No coefficient depends on deciding whether K215 halts. The 430 machine transitions, 943 tiles, masks, weighted quartic coefficients and all fixed powers are bounded, explicitly reconstructible data. Auxiliary names inside the displayed formulas denote polynomial expressions, not additional existential variables. Fixed sums with large numerical limits are ordinary polynomials; their efficient evaluators are supplied.

1.3 Verification boundary

The archive separates three kinds of evidence. First are the inherited semantic results, pinned source proofs and the existing Lean endpoint. Second are the new ordinary mathematical proofs: the coefficient adapter, square and divisibility mergers, the cofinal Jones restriction, and the algebraic refactorings. Third are executable finite checks of coefficient bytes, symbolic identities, degree witnesses and circuit transcription. The finite checks are useful for catching errors in a concrete equation; they do not prove universal compiler correctness by testing examples.

The new work is not an end-to-end Lean or Rocq development. In particular, the ten-variable route imports a theorem from the explicitly identified 2026 preprint [11]; the sixteen-variable route independently uses the Jones reduction and does not depend on that preprint. The ordinary proofs use finite arithmetic, induction, elementary algebra and the cited Diophantine theorems, all available within ZFC. No truth or consistency assumption about ZFC is needed to prove the biconditional.

2 A smaller complete coefficient tuple

The mixed-radix clause encoding of package 25 and the arbitrary-odd-factor rank controller of package 24 combine without changing the 943-tile language. This removes two independent

power-of-five roundings:

$$\rho = 2^{121}, \qquad L = 8801, \qquad d = 121 \cdot 8801 = 1064921, \qquad B = 2^d.$$

The largest fixed port has 1,070,004 bits, compared with 1,958,376 bits in packages 25 and 27. The four raw payloads occupy 533,099 bytes instead of 977,220 bytes. These reductions are approximately 45.36% and 45.45%, respectively. Variables, degree and arithmetic gate count of a compatible skeleton do not change.

2.1 The literal bounded recipe

Retain the ordered 943 tiles in the bundled literal table, in slot order west, north, east, south. Start is tile 0 and the impossible tile is tile 1. The new builder reconstructs this ordered table from all 430 literal K215 transitions. Put

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1032\} \\ & \cup \{1063, 1064, 1095, 1096, 2195, 3296\}.\end{aligned}$$

The four copy offsets are $1023, 1013, 1003, 983$; the four anchors are $1095, 1096, 2195, 3296$. Set $W_j = 2^{10+2j}$ for $0 \le j < 44$. If tile i has edge labels x_{it}, define

$$c_i = 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{it} + 512) W_{10t+j} + \sum_{j=40}^{43} W_j.$$

At copy exponent offset$_t$ $+j$, set $c_e = W_{10t+j}$. The four anchors receive $W_{40}, \dots, W_{43}$ in the listed order; $c_{1063} = c_{1064} = 0$. Put

$$\mu = 1022 + \sum_{j=0}^{43} W_j, \qquad Z = (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799}.$$

The exact constants are

$$\begin{aligned} D_C &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2200} + \sum_{e \in \mathcal{E}} c_e (\rho^{4393-e} + \rho^{5495-e}), \\ D_R &= \rho^{42}, \\ M_C &= B - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \ne 1}} \rho^e - 2\rho^{1063}, \end{aligned}$$

$$M_{F,0} = 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1013+j} + \rho^{1065+j}) + \rho^{2195} + \rho^{3296} + Z.$$

The four ports, always replaced together, are

$$\boxed{\beta = B - 1, \quad K = D_C + BD_R, \quad M_C, \quad M_F = M_{F,0} + B - 1.} \tag{2.1}$$

The last port is already shifted. Every number in this definition is computed by a bounded finite recipe, with no search for a halting run.

2.2 What the finite geometry certifies

The clause coefficients and copy/anchor geometry are unchanged from package 25. If n selectors are active, then $0 \le n \le 943$ and the low ten bits of the clause sum equal n. The mask 1022 forces $n = 0$ or 1, without any assumption about carry freedom in higher fields. Once this is

known, every radix-four field is a sum of two Boolean values. Its masked low bit forces the copy or anchor to equal the selected value.

The new cell circumference satisfies the exact identities

$$\{(4393 - e) \bmod 8801 : e \in \mathcal{E}\} \cap \{(5495 - e) \bmod 8801 : e \in \mathcal{E}\} = \varnothing,$$
$$\{(2195 - e) \bmod 8801 : e \in \mathcal{E}\} \cap \{(3296 - e) \bmod 8801 : e \in \mathcal{E}\} = \{0\}.$$

Thus there is a clean clause band under every rotation, and the two occupied anchors force whole-cell alignment. The ordinary overlap identities are the same as in package 25: targets $1013 + j$ compare sources $983 + j$ and $1013 + j$, while targets $1065 + j$ compare $1003 + j$ and $1023 + j$; the anchor targets compare 1095 with 2195 and 1096 with 3296. The builder checks all 69,230 scalar basis identities. Linearity extends them to every native assignment.

The uniform pre-decoding bound is

$$\mathcal{M} = (989 + 2)(2 \sum_{e} c_e + 6), \qquad T_0 = \max(4\mathcal{M} + 8, 2\mu + 4, 16).$$

Exact arithmetic gives $2^{119} < T_0 < 2^{120} < \rho$. Hence 121 is the least odd radix width for this unchanged bound. This is not a global minimum. The maximum unshifted product exponent is $5495 + 3296 = 8791$, so all eight padding cells are identically zero in canonical assignments, including after every allowed dummy change. Their additional tests preserve completeness and may be ignored in soundness. Also $2(2^{119} - 1) + 4 < \rho$, so the mask fits the inherited slack estimate.

The padding contributes $7 \cdot 119 + 27 = 860$ bits; the core high mask has $2 \cdot 53 + 20 + 2 + 1 = 129$ bits. Materialization confirms

$$\operatorname{popcount}(M_{F,0}) = 989, \qquad \operatorname{popcount}(M_C) = d - 989,$$
$$0 < M_C, M_{F,0} < B - 1, \qquad M_C \equiv 2 \pmod 4, \qquad M_{F,0} \equiv 4 \pmod 8.$$

2.3 An explicit controller for the actual packed rank

The new d is not a power of five, so the old five-adic tuning statement cannot simply be reused. Factor $d = 11^2 \cdot 13 \cdot 677$ and set

$$\lambda = 2341, \qquad k = 2340 = \lambda - 1.$$

Trial division proves that λ is prime. Exact modular arithmetic gives the following orders:

Modulus	121	13	677	2341
Order of B	10	12	52	60

Thus $\operatorname{ord}_d(B) = 780 \mid k$, and

$$B^{2340} \bmod 2341^2 = 2556373 = 1 + 1092 \cdot 2341, \qquad v_\lambda(B^k - 1) = 1.$$

For $H_0 = 1 + BD_C + B^2(D_R + 1)$ the necessary unit residues are

Prime p	$B \bmod p$	$(B^2 - 1) \bmod p$	$H_0 \bmod p$
11	2	3	9
13	6	9	7
677	611	293	654
2341	2100	1896	1225

Choose sufficiently large independent powers of λ for tableau width w and height h, and put $N = wh = \lambda^n$, $q = B^N$. The inherited language allows arbitrary sufficiently large independent rectangular padding. Impose $N > d\lambda^2$ and $h > \lambda$. The order table gives $q \equiv B^w \equiv B \pmod{d\lambda}$. Activating the lower dummy in cell j changes the actual packed Pell index by the inherited exact formula

$$\delta R = -\Gamma B^j, \qquad \Gamma = \rho^{1063}(q^2 - 1)[1 + q(D_C + BD_R + B^w)].$$

The unit table proves $\gcd(\Gamma, dN) = 1$.

Boolean coverage. Set $T = N/\lambda$ and $C_0 = d\lambda$. Every residue modulo $dN = C_0T$ is a Boolean subset sum of

$$\{B^{2+kj} : 0 \le j < T\} \cup \{B^3\}.$$

Indeed, binomial lifting and $B^k \equiv 1 \pmod{d}$ show that its first T powers are exactly the residues $1 + C_0z$, z modulo T. Normalize the desired residue by B^{-2} and choose $\varepsilon \in \{0, 1\}$ so that subtracting εB leaves a nonzero residue modulo λ. Let a be the least residue modulo C_0. Then $1 \le a < C_0 < T$ and $\gcd(a, T) = 1$. Write the target as $a + C_0u$, solve

$$az_0 + \frac{a(a-1)}{2} \equiv u \pmod{T},$$

and select the a distinct orbit entries whose additive coordinates are $z_0, \dots, z_0 + a - 1$. Include B^3 exactly when $\varepsilon = 1$. This constructs the required subset with no repeated position. The largest cell index is

$$2 + (\lambda - 1)(N/\lambda - 1) = N - N/\lambda - \lambda + 3.$$

Since $w < N/\lambda$, its translates by one and by w do not wrap. The extra position 3 is distinct because $k > 1$.

Start with zero control bits and actual rank R_0. Applying coverage to $(R_0 - dw)\Gamma^{-1}$ tunes $R \equiv dw \pmod{dN}$, exactly the temporal rotation required by $X = 2^R$. The previously proved carry, slack and padding bounds hold under every selected change. This proves unbounded completeness; the finite modular certificates alone would not do so.

2.4 Full interface and verification boundary

All of b, L, d, N are odd. With $t = dN$, the canonical Start and low-mask congruences still give $R \equiv 3 \pmod 4$, while the exact mask population identity gives $\mathrm{popcount}(R) = 3t + 2$. Therefore $X/q = 2^{R-t}$ is a positive square. The inherited even main-Pell parameter and strong auxiliary index preserve the odd positive multiplier; the half-binomial valuation gives $v_2(Y) = 3t$, so $s = Y/q^3$ remains odd. These are precisely the extra completeness promises needed for the compatible signed reductions.

Consequently the complete tuple (2.1) satisfies the same blank-K215 native interface as the source tuple, including its positive slack and signed parity requirements. Composing this ordinary adapter proof with a compatible arithmetic-skeleton proof and the K215 endpoint preserves the ZFC-provable equivalence between solvability and existence of a finite ZFC refutation.

The bundled standard-library builder reconstructs the exact payloads, tests all finite geometry identities and modular certificates, and records hashes and bit lengths. It does not decide consistency or supply an end-to-end Lean/Rocq certificate. The complete new adapter proof is also provided separately, with the relevant pinned source proofs and notices.

3 The common positive quartic and its literal index

We use the unshifted positive quartic of package 30 as the coding source. This choice keeps the inherited coefficient index unchanged as a symbolic recipe. The cheaper 60-operation system later in this report is an equivalent presentation; it is not silently substituted into the index compiler.

Let β, K_0, M_C, M_F denote the complete tuple of (2.1); K_0 is that equation's port K and is distinct from subsequent polynomial expressions also traditionally called K. The 31 positive coordinates, in their fixed order, are

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma, N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta).$$

Put

$$q = \beta J + 1, \quad k = \eta + \zeta, \quad H = 4a + 3, \quad C = q - F - \alpha.$$

The parent is the sum of the squares of the residuals of these 22 equations, with "left minus right" as the residual convention:

$$\begin{aligned}
L &= q^2 - 1, & N &= q(L+1),\\
X &= wq, & a &= E + sN,\\
E &= X(a - E), & c &= k(a - E) + \eta,\\
P &= E(c - \eta) + k, & \tau^2 &= P(P - k) + 1,\\
D &= X + ac + \gamma H, & \Delta &= a^2 + H,\\
C_2 &= c^2, & D^2 &= \Delta C_2 + 1,\\
k &= jc - V + 1 + hE, & F_2 &= f^2,\\
V &= of - c, & g &= y^2 - V^2,\\
I &= iC_2, & B_{\mathrm{aux}} &= I^2,\\
B_{\mathrm{aux}} &= \Delta(F_2 - 1), & y^2 - B_{\mathrm{aux}} g &= 1,
\end{aligned} \tag{3.1}$$

together with the two transport equations

$$jc - V = (q - F)(N - q) - CL + (M_C + qM_F)J, \tag{3.2}$$

$$(K_0 + w)C + q - F - z(q - 1) = 1. \tag{3.3}$$

The order in the literal JSON, rather than the grouping on this page, is authoritative for the coefficient indices. There are no products of more than two coordinates in a residual after the displayed abbreviations are substituted: every residual has total degree at most two.

Write this sum of squares as Q_{31} and let

$$P_\star(x_0, x_1, \ldots, x_{31}) = Q_{31}(x_1 + 1, \ldots, x_{31} + 1).$$

The dummy input x_0 does not occur. The coefficient of τ^4 in Q_{31} is 1, so the degree is exactly four. At the all-zero nonnegative input the residual $E - X(a - E)$ is 1, hence $P_\star(a, 0, \ldots, 0) \geq 1$ for every integer a. Thus Sun's source construction applies directly; there is no need to append a variable or square this quartic again.

Proposition 3.1. *With the complete tuple (2.1), ZFC proves*

$$(\exists \boldsymbol{n} \in \mathbb{N}^{31}\ P_\star(0, \boldsymbol{n}) = 0) \iff (\textit{there is a finite ZFC refutation}).$$

Proof. Translation by 1 is a bijection from $\mathbb{N}$ to $\mathbb{Z}_{>0}$. A sum of integer squares is zero exactly when every residual is zero. The package-30 parent proof identifies these positive solutions with the compatible native arithmetic skeleton. Section 2 supplies that skeleton with a complete tuple preserving its soundness, completeness, slack and parity interface. The native compiler and the K215 endpoint then give the assertion. The strong auxiliary Pell index must be converted to the ordinary index as in the inherited parent proof; the tuple replacement does not change that conversion. □

3.1 Weighted coefficients with no unspecified index

Put $\nu = 31$ and $L_\star = 5^{32} = 23283064365386962890625$. Write

$$P_\star = \sum_{|\boldsymbol{a}|\le 4} p_{\boldsymbol{a}} x_0^{a_0} \cdots x_{31}^{a_{31}}, \qquad c_{\boldsymbol{a}} = (4 - |\boldsymbol{a}|)! \prod_{j=0}^{31} a_j!\, p_{\boldsymbol{a}}.$$

Define $e(\boldsymbol{a}) = \sum_{j=0}^{31} a_j 5^j$. All $58,905 = \binom{36}{4}$ positions of total weight at most four are distinct, and $e(\boldsymbol{a}) < L_\star$. There are 402 nonzero weighted coefficient records when the four ports are treated as fixed symbolic parameters. The file `signed16/data/constants.json` gives all of them as explicit finite recipes. The verifier expands the normalized parent symbolically and checks every record, including factorial weights.

For the new tuple, the explicitly materialized integer

$$z_\star = 2^{4269855}$$

satisfies

$$|c_{\boldsymbol{a}}| + 1 < z_\star \quad \text{for every } |\boldsymbol{a}| \le 4. \tag{3.4}$$

The check uses exact integer arithmetic on the actual port payloads, not a floating-point estimate. This also bounds every $|p_{\boldsymbol{a}}|$ by $z_\star$, since its factorial multiplier is a positive integer.

For the Jones route define the fixed polynomial and fixed integers

$$e_\star(T) = \sum_{|\boldsymbol{a}|\le 4} (z_\star + c_{\boldsymbol{a}}) T^{L_\star - e(\boldsymbol{a})},$$
$$y_\star = e_\star(2z_\star), \qquad \kappa = (2z_\star)^{L_\star+1} = 2^{9941533207893371582031676985 6}.$$

Every digit lies strictly between 0 and $2z_\star$ and its position is distinct, so $0 < y_\star < \kappa$. These are bounded definitions of actual integers. The huge $y_\star$ need not be a leaf of a circuit, and the power-of-two recipe for κ is explicit even though expanding its binary string is not practical.

For the ten-variable route we instead use the sparse polynomial

$$D_\star(T) = \sum_{|\boldsymbol{a}|\le 4} c_{\boldsymbol{a}} T^{L_\star - e(\boldsymbol{a})}.$$

It has exactly the same weighted coefficient data, without the all-position $z_\star$ padding. This is one reason to preserve the literal parent and its coordinate order across the two independent reductions.

4 An explicit equation in ten unrestricted integers

The smallest unknown count in this package is achieved by the following fully fixed equation:

$$F_{10}(f, g, h, k, l, w, x, y, z, m) = 0, \qquad (f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}.$$

Its division-free circuit uses **1,442 binary arithmetic operations** and has exact total degree

$$126496 \cdot 5^{31} + 27064 = 589042901992797851562527064.$$

The general ten-integer reduction is due to Sun's 2026 preprint; this report does not claim a new universal unknown-count theorem. Our contribution here is its completely specified Con(ZFC) specialization, together with three local reductions of its degree and arithmetic circuit. The two primary sources are Sun's 2021 paper, especially the explicit proof of Theorem 3.1, and his 2026 preprint, Sections 3, 6 and 7.[1] The literal Sun final formula is retained as a reference variant.

4.1 Fixing every source coefficient

Let $r_1, \dots, r_{22}$ denote the quadratic residuals of the positive 31-variable system, with the complete new four-port tuple. Define

$$P_0(n_1, \dots, n_{31}) = \sum_{j=1}^{22} r_j(n_1 + 1, \dots, n_{31} + 1)^2.$$

The preceding reduction proves that a nonnegative zero of P_0 exists exactly when ZFC is inconsistent. Its degree is four, and $P_0(0) > 0$: the fifth residual $E - X(a - E)$ equals one at the all-one parent assignment. Apply Sun's source construction to

$$P(a, n_1, \dots, n_{31}) = P_0(n_1, \dots, n_{31}), \qquad a = 0.$$

The input a is dummy. This direct specialization avoids any unspecified index for a recursively enumerable set.

For $\alpha \in \mathbb{N}^{32}$ with $|\alpha| \le 4$, write p_α for the corresponding ordinary coefficient of P and set

$$c_\alpha = (4 - |\alpha|)! \prod_{j=0}^{31} \alpha_j!\, p_\alpha, \qquad e(\alpha) = \sum_{j=0}^{31} \alpha_j 5^j.$$

Exactly 402 nonzero weighted coefficient recipes are retained in `signed10/data/constants.json`; unlisted coefficients are zero. The verifier reconstructs all of them from the shifted residual squares. All nonzero records have $\alpha_0 = 0$. The checked coefficient bound is

$$|c_\alpha| + 1 < 2^{4269855}.$$

Fix

$$\beta_S = 2^{4269882} = 2^{2 \cdot 2134941}.$$

Because $48 \cdot 33^4 = 56924208 < 2^{27}$ and $|p_\alpha| \le |c_\alpha|$,

$$\beta_S > 33^4\, 4!\, 2 \max_\alpha |p_\alpha|.$$

This is precisely the source bound in Sun 2021 for $\nu = 31$, $\delta = 4$ and $p = 2$. The additional constant is supplied both by its exact power and by a compressed binary payload. The binding of every fixed leaf is `signed10/data/INSTANCE.json`.

[1] Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Sci. China Math. 64 (2021), 281–306, https://arxiv.org/html/1704.03504v7; Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, 2026 preprint, https://arxiv.org/html/2609.01594v2.

4.2 The complete source polynomials

Put $t = 5^{31}$ and $L_c = 5^{32} = 5t$. Define

$$
\begin{aligned}
b &= 1 + 3f, & \mathcal{B} &= \beta_S b^4,\\
\ell &= \sum_{j=1}^{31} \mathcal{B}^{5^j}, & D_c &= \sum_{|\alpha|\le 4} c_\alpha \mathcal{B}^{L_c - e(\alpha)},\\
N_0 &= \mathcal{B}^{t+1}, & N_1 &= 4\mathcal{B}^{9t+1},\\
N &= N_0 N_1, & M_c &= N_0 - 1 - (b-1)\ell,\\
T_c &= M_c + (\mathcal{B}-2)\mathcal{B}^{L_c} N_0, & J_c &= 2(1+g)^4 D_c + \sum_{i=0}^{9t} \mathcal{B}^{i+1},\\
S_c &= g + J_c N_0, & R_c &= (S_c + T_c + 1)N + T_c + 1,\\
X &= (N-1)R_c, & Y &= N^2,\\
\mathscr{C} &= b\mathcal{B}^t.
\end{aligned}
$$

All powers and summation limits are fixed integers. The apparent division by $p-1$ in Sun's general formula disappears at $p = 2$. The mask M_c is his digit mask in telescoped form. The bound has the prescribed form $\mathscr{C} = 2^{4269882t} b^{4t+1}$.

Sun's source theorem supplies, from a nonnegative zero of P_0, witnesses with $f > 0$, b a square and a power of two, $b \le g < \mathscr{C}$, and $Y \mid \binom{2X}{X}$. Conversely, $f \ne 0$, $0 \le g < 2\mathscr{C}$, the same square and power conditions on b, and that divisibility imply a nonnegative zero of P_0. The construction also gives the strict size bounds $X > 3b$ and $Y > \max\{b, 256\}$ required by Sun 2026. Thus all source hypotheses are checked for this particular, fully fixed polynomial.

4.3 The bridge and the shared Lucas witnesses

Use the following polynomial abbreviations, in the displayed order:

$$
\begin{aligned}
L &= lY, & U &= 2LX, & V &= 4gwY, & W &= bw,\\
A &= U(V+1), & B &= 2X+1, & C &= B + (A-2)h,\\
P_n &= U^2 V, & K &= X + 1 + k(P_n - 2),\\
S &= 2A - 5, & T &= 3WC - 2(W^2 - 1),\\
R_2 &= (P_n^2 - 4)K^2 + 4.
\end{aligned}
$$

Then set

$$
\begin{aligned}
\Delta &= 8\mathscr{C}^3 g K^2 - g^2\big(32(C - KL)^2\mathscr{C}^3 + g^2 K^2\big),\\
O &= f^2 l^2 \Delta, \qquad \xi = zx,\\
D &= (A^2 - 4)C^2 + 4, \qquad E = C^2 D\xi,\\
F &= 4(A^2 - 4)E^2 + 1,\\
G^* &= 2(A+2)(A-2)^2 E^2 - 1, \qquad \Omega = 16E,\\
H^* &= C + (B - C)F + \Omega F y,\\
I^* &= ((G^*)^2 - 1)(H^*)^2 + 1.
\end{aligned}
$$

These are Sun 2026's formulas (6.1)–(6.7), followed by his modified Lucas certificate (3.1)–(3.7). The symbol P_n only factors a repeated product. The integer S is always nonzero and odd.

His positive bridge supplies b square, R_2 square, $S \mid T$, $C = u_B(A)$, and $O > 0$. In fact $O \geq 4$: on this branch $g \geq b \geq 4$, and $\Delta > 0$ is a multiple of g; also $f, l \neq 0$. His reverse bridge states that

$$b \text{ square}, \quad O > 0, \quad \xi \neq 0, \quad DFI^* \text{ square}, \quad R_2 \text{ square}, \quad S \mid T$$

imply a nonnegative zero of P_0. These precise bridge statements are the external Lucas-theoretic inputs used below.

4.4 Three local simplifications

A quintic merger. Use

$$\pi(v) = (v^3 - v - 1)(v^2 + 23), \qquad \Sigma = \xi S^5,$$

and

$$r = T + Sy, \qquad \Theta = (r^3 - rS^2 - S^3)(r^2 + 23S^2).$$

The polynomial π has a root modulo every positive integer. For an odd prime other than 23, either its cubic factor has a simple root, or that cubic is irreducible and its discriminant -23 is a square: the Frobenius permutation is an even 3-cycle. In the latter case the quadratic factor has a simple root. At 23 the cubic has the simple root 3. At 2, $-23 \equiv 1 \pmod 8$ has a 2-adic square root. Hensel lifting and the Chinese remainder theorem finish the argument. Neither factor has a rational root.

If $S \mid T$ and $\xi \neq 0$, prescribe a residue class of y modulo ξ so that $\Sigma \mid \Theta$; Sun's full residue interpolation for the Lucas witness allows exactly this prescription. Conversely, if a prime has $\nu_p(S) > \nu_p(T)$, the two factors of Θ have valuations $3\nu_p(T)$ and $2\nu_p(T)$, contradicting $S^5 \mid \Theta$. Hence $\Sigma \mid \Theta$ forces $S \mid T$. Moreover $\Theta \neq 0$ whenever $S \neq 0$, by the absence of rational roots. This replaces Sun's degree-six merger with a degree-five one.

One square in place of two. Define

$$r_0 = b(1 + (3O - 4)b^2z^2), \qquad r_1 = R_2, \qquad r_2 = DFI^*.$$

For $z \neq 0$, a square r_0 forces b to be a positive square and $O > 0$. Indeed its two factors are coprime. The congruence $b \equiv 1 \pmod 3$ excludes zero and negative squares. A zero factor would force $b = 1$, hence $f = O = 0$ and $1 - 4z^2 = 0$, which is impossible. The remaining factor is a square of the form $1 + (3O - 4)(bz)^2$ with $bz \neq 0$, so the standard Pell positivity lemma gives $O > 0$.

For completeness, the positive bridge has $O \geq 4$, so $d = 3O - 4 > 0$ is nonsquare because it is 2 modulo 3. A positive Pell solution $u^2 - dv^2 = 1$ can be chosen with $b \mid v$: raise a fundamental Pell unit to the order of its image modulo b. Taking $z = v/b > 0$ makes r_0 a square. Sun's interpolation theorem then chooses ξ divisible by this z, so $x = \xi/z$ is still an integer. No witness is added.

Smaller separating weights. For the actual polynomial r_0, $z \neq 0$ implies $|r_0| > 1$. If $r_0 = 0$ or $r_0 = \pm 1$, its factorization forces $b = 1$, hence $f = O = 0$; but then $r_0 = 1 - 4z^2 \leq -3$, a contradiction. Also

$$\sqrt{|r_1|} \leq P_n^2 + K^2 + 2,$$

because $|r_1| \leq (P_n^2 + 4)K^2 + 4$ and the difference from the square of the right-hand side is $P_n^4 + K^4 + P_n^2K^2 + 4P_n^2 \geq 0$. Thus the possibly signed weights

$$w_0 = 1, \qquad w_1 = r_0, \qquad w_2 = r_0(P_n^2 + K^2 + 3)$$

satisfy

$$|w_1| > \sqrt{|r_0|}, \qquad |w_2| > \sqrt{|r_0|} + |w_1|\sqrt{|r_1|}.$$

If a signed sum of these weighted roots is rational, any Galois automorphism changing a root has a largest changed index. Its term has magnitude at least the corresponding weight and cannot be cancelled by the earlier terms. Hence every root is rational, and the integer radicands are squares. The same argument covers negative radicands by complex absolute values, and zero radicands contribute no term.

4.5 The final polynomial, without radicals or division

Set

$$X_n = \Sigma m - \Theta, \qquad A_i = \Sigma^2 w_i^2 r_i \quad (0 \le i \le 2),$$

and then

$$V_n = X_n^2 + A_2 - A_0 - A_1, \qquad W_n = V_n^2 + 4X_n^2 A_2 - 4A_0 A_1.$$

The requested equation is

$$\boxed{F_{10} = W_n^2 - 16V_n^2 X_n^2 A_2 = 0.} \tag{4.1}$$

All preceding letters are polynomial abbreviations, not additional unknowns. Its formal norm identity is

$$F_{10} = \prod_{\varepsilon \in \{\pm 1\}^3} \left(X_n + \sum_{i=0}^{2} \varepsilon_i \sqrt{A_i} \right).$$

This identity explains the construction; equation (4.1) itself contains no radicals. Its 35-term formal expansion is included in the archive and checked exactly against the eight signed factors.

For a nonnegative zero of P_0, the positive bridge, the merged-square Pell construction and the full residue-interpolation theorem supply r_0, r_1, r_2 square, $\Sigma \neq 0$ and $\Sigma \mid \Theta$. Choose the integer m to make a norm factor vanish. Conversely, if $F_{10} = 0$, then $\Sigma = 0$ would give $F_{10} = \Theta^8 \neq 0$. Thus $\xi = zx \neq 0$ and $z \neq 0$, so the structural separating-weight argument applies. A vanishing norm factor forces the three radicands to be squares and $\Sigma \mid \Theta$. The quintic merger gives $S \mid T$ and the coprime square gives b square and $O > 0$. Sun's reverse bridge gives a nonnegative zero of P_0.

Combining this with the fixed parent/compiler endpoint yields

$$\mathrm{ZFC} \vdash \left[\neg\exists (f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10} \ F_{10} = 0 \right] \longleftrightarrow \mathrm{Con(ZFC)}.$$

This invokes the stated mathematical theorems, including Sun's 2026 preprint. The computational receipt verifies finite coefficient and circuit claims; it is not an end-to-end formal proof of this equivalence.

4.6 Reproduction and quantitative comparison

The two commands `python build_signed10.py` and `python verify_signed10.py`, run in the `signed10` directory, use only the Python standard library. The source sum is evaluated by a shared-power Horner circuit; the geometric sum uses repeated doubling, without division. All constant powers are expanded into binary gates. The main circuit has 566 additions, 25 subtractions and 851 multiplications, with 79 fixed leaves. A nonzero leading coefficient after diagonal specialization modulo 101, 1009 and 1000003 proves that the circuit degree upper bound is attained.

For transparency, the main degree calculation is as follows. Here $t = 5^{31}$, and X_n, A_i refer to the finalizer rather than the source polynomial X:

Expression	Total degree
X	$120t + 24$
Y	$80t + 16$
S	$280t + 59$
r_0	$1132t + 250$
$r_1 = R_2$	$1920t + 402$
$r_2 = DFI^*$	$24640t + 5268$
Σ	$1400t + 297$
X_n	$1400t + 310$
A_0	$3932t + 844$
A_1	$6984t + 1496$
A_2	$31624t + 6766$

The degree of A_2 strictly exceeds those of X_n^2, A_0, A_1. Consequently the top part of the final norm is A_2^4, giving $\deg F_{10} = 126496t + 27064$. The recorded nonzero diagonal coefficients are exact finite certificates that these top parts do not cancel for the actual fixed coefficient tuple.

Variant	Operations	Exact degree, $t = 5^{31}$
Sun's four-square sextic specialization	1492	$2407680t + 514640$
Four squares, quintic merger	1492	$2403200t + 513696$
Three squares, Sun's generic weights	1445	$807360t + 172560$
Three squares, universal successive weights	1444	$158592t + 33880$
Three squares, structural signed weights	1442	$126496t + 27064$

All five variants have the same ten unrestricted integer unknowns and the same fixed source coefficients. The complete proof, exact circuits, coefficient recipes, binary payloads, source references and verification receipts are in the `signed10` directory.

5 A sixteen-integer equation with a smaller arithmetic circuit

This section gives an explicit intermediate tradeoff between the direct Pell/compiler equations and the smaller-variable construction elsewhere in this report. The equation has sixteen unrestricted integer unknowns, degree

$$813056 \cdot 5^{32} + 10016 = 189304351806640625000000010016,$$

and an evaluator with 1617 binary arithmetic operations. It combines package 30's 31-variable quartic, package 29's signed Jones interface and relation-combining construction, and package 27's digit-polynomial evaluation identities [5, 6, 7]. The new steps are a cofinal restriction of the coding base, the removal of its sign guard, smaller root-separation weights, and an automatic nonzero condition. No minimum-variable or global operation-minimality claim is made.

Every abbreviation below denotes a polynomial expression, not another quantified unknown. Every exponent in the equation is a fixed natural number. The large fixed coefficients are defined by finite recipes; they are not existential parameters.

5.1 The fixed instance and the upper-bound coefficient

Let Q_{31} be the sum of the 22 squared quadratic residuals in the *unshifted-transport* parent `signed16/data/parent_quartic31.json`. This is the package-30 parent, with its whole four-port compiler tuple instantiated by the 121-by-8801 construction of this report. In particular, this

parent is not silently replaced by the locally shifted quartic evaluator. The coefficient interface is the same polynomial interface in (β, K_0, M_C, M_F); all four ports use this one complete tuple.

Put $\nu = 31$, $L = 5^{32}$, and define the normalized polynomial

$$\mathcal{P}(x, n_1, \ldots, n_\nu) = Q_{31}(n_1 + 1, \ldots, n_\nu + 1).$$

The dummy input x is unused. The degree is four, and $\mathcal{P}(x, 0, \ldots, 0) > 0$: the fifth parent residual is 1 when all positive parent coordinates equal 1. Write

$$\mathcal{P}(x, n_1, \ldots, n_\nu) = \sum_{\substack{\alpha \in \mathbb{N}^{\nu+1} \\ |\alpha| \le 4}} p_\alpha(x, n_1, \ldots, n_\nu)^\alpha,$$

$$c_\alpha = (4 - |\alpha|)! \left(\prod_{j=0}^{\nu} \alpha_j! \right) p_\alpha, \qquad \exp(\alpha) = \sum_{j=0}^{\nu} \alpha_j 5^j.$$

There are 402 nonzero symbolic weighted-coefficient records. The omitted records mean $c_\alpha = 0$, not omitted baseline digits. The complete finite coefficient bounds supplied with the new tuple establish

$$|c_\alpha| + 1 < 2^{4269855} \qquad (|\alpha| \le 4).$$

Set

$$z_* = 2^{4269855}, \qquad e_*(T) = \sum_{|\alpha| \le 4} (z_* + c_\alpha) T^{L - \exp(\alpha)},$$

$$\upsilon_* = \sum_{j=1}^{\nu} (2z_*)^{5^j}, \qquad y_* = e_*(2z_*), \qquad \kappa = (2z_*)^{L+1}. \tag{5.1}$$

Thus, exactly,

$$\kappa = 2^{4269856(5^{32}+1)} = 2^{99415332078933715820316769856}.$$

The final polynomial uses z_*, κ, and the weighted coefficients as fixed leaves. Neither y_* nor υ_* is a variable or a leaf needed by its evaluator.

Lemma 5.1 (The fixed upper bound)**.** *The recipes in* (5.1) *define positive integers and* $y_* < \kappa$*. They give the Jones index for the normalized parent* $\mathcal{P}$ *with parameter* $\nu = 31$*.*

Proof. Each digit $z_* + c_\alpha$ lies strictly between 1 and $2z_*$. Since every $\alpha_j \le 4$, uniqueness of base-five expansions makes the positions $\exp(\alpha)$ distinct. Also $\exp(\alpha) \le 4 \cdot 5^\nu < L$. Consequently y_* is a base-$2z_*$ integer whose positions lie in $\{0, \ldots, L\}$, with each digit less than $2z_*$. Therefore

$$0 < y_* \le (2z_* - 1) \sum_{r=0}^{L} (2z_*)^r = (2z_*)^{L+1} - 1 < \kappa.$$

The coefficient inequality, the power-of-two choice of z_*, and the displayed definitions of υ_*, y_* are exactly the admissible index conditions used by the inherited parametric Jones theorem. This uses that theorem at $\nu = 31$, not the convenience existence theorem whose internal parent has 58 variables. □

5.2 The literal sixteen-coordinate equation

The unrestricted integer coordinates, in the emitted order, are

$$t, g_0, h_0, s_0, w_0, \phi, i_0, j_0, \quad z_1, z_2, z_3, z_4, z_5, z_6, \quad v, u. \tag{5.2}$$

First define

$$b=\kappa t^2,\qquad g=g_0+1,\quad h=h_0+1,\quad s=s_0+1,\quad w=w_0+1,\quad i=i_0+1,\quad j=j_0+1,$$

and the fixed-degree coding polynomials

$$\mathcal{B}=b^5,\quad q=\mathcal{B}^L,\quad \ell=\sum_{r=1}^{\nu}\mathcal{B}^{5^r},\quad \lambda=\sum_{r=0}^{4L-1}\mathcal{B}^r,\quad e=e_*(\mathcal{B}),\quad \theta=\mathcal{B}-2z_*,$$
$$\mathcal{N}=q^{16},\qquad G_{\mathrm{cap}}=32b\mathcal{B}^{5^\nu}. \tag{5.3}$$

The packed rank is

$$\begin{aligned}R=&\left[g+eq^3+\ell q^5+\left\{2(e-z_*\lambda)(1+\mathcal{B}+g)^4+\lambda\mathcal{B}+\lambda\mathcal{B}q^4\right\}q^7\right](\mathcal{N}^2-\mathcal{N})\\&+\left[q^3-b\ell+\ell+\theta\lambda q^3+(\mathcal{B}-2)q^8\right](\mathcal{N}^2-1).\end{aligned} \tag{5.4}$$

Continue with the unchanged Jones ratio expressions:

$$\begin{aligned}&U=\mathcal{N}^2w, && Y=\mathcal{N}^2s, && M=RY, && P=2M^2U,\\&K=R+1+h(P-1), && A=M(U+1), && B_{\mathrm{J}}=2R+1, && W_{\mathrm{J}}=bw,\\&C=B_{\mathrm{J}}+W_{\mathrm{J}}+\phi, && J=4A-5, && Q=3W_{\mathrm{J}}C-2(W_{\mathrm{J}}^2-1),\\&D=(A^2-1)C^2+1, && E=iJDC^2, && F=(A^2-1)E^2+1,\\&G_1=A+F(CD+1-A), && H=B_{\mathrm{J}}+jC, && I=(G_1^2-1)H^2+1.\end{aligned} \tag{5.5}$$

Finally put

$$\begin{aligned}&a_0=(P^2-1)K^2+1, && a_1=DFI,\\&m=K^2G_{\mathrm{cap}}-4(C-KY)^2G_{\mathrm{cap}}-8gK^2, && D_0=JF, && T_0=QF+J(H-C).\end{aligned} \tag{5.6}$$

The auxiliary letter m denotes the margin in this section.

The five base coordinates that retain explicit sign guards are

$$(x_1,x_2,x_3,x_4,x_5)=(g_0,s_0,w_0,\phi,j_0).$$

Starting with $C_0=1$, form the six-stage stack

$$\begin{aligned}&d_r=3x_r+2, && B_r=1+4d_rC_{r-1}^2z_r^2, && C_r=C_{r-1}B_r && (1\le r\le 5),\\&d_6=3m-1, && B_6=1+4d_6C_5^2z_6^2, && C_*=C_6=C_5B_6.\end{aligned} \tag{5.7}$$

The circuit names the stack outputs `guard1`, …, `guard6`; the ratio-block expression $B_{\mathrm{J}}=2R+1$ is separate.

For arbitrary integer arguments define

$$\mathcal{R}(D',T',v)=\big((T'+D'v)^3-D'^2(T'+D'v)-D'^3\big)\big((T'+D'v)^2+23D'^2\big). \tag{5.8}$$

Using this polynomial, set

$$\begin{aligned}&Z=i\prod_{r=1}^{6}z_r, && S=ZD_0^5, && T=\mathcal{R}(D_0,T_0,v),\\&w_1=C_*, && V_0=P^2+K^2+2, && w_2=C_*V_0,\\&X=Su-T, && A_0=S^2C_*, && A_1=S^2w_1^2a_0, && A_2=S^2w_2^2a_1,\\&V=X^2+A_2-A_0-A_1, && W=V^2+4X^2A_2-4A_0A_1.\end{aligned} \tag{5.9}$$

The promised ordinary integer polynomial is

$$\boxed{F_{16}=W^2-16V^2X^2A_2.} \tag{5.10}$$

In particular, t is deliberately absent from Z. Its nonvanishing will follow from the recovered margin; omitting it saves one multiplication.

5.3 The exact imported interface and the new cofinal restriction

For an integer a, write $\mathrm{Sq}(a)$ for the assertion $\exists r \in \mathbb{Z}\ (a = r^2)$. The inherited Jones ratio equivalence applies to the eight *nonnegative* coordinates

$$\varepsilon, g_0, h_0, s_0, w_0, \phi, i_0, j_0, \qquad b = y_* + 1 + \varepsilon.$$

For the fixed normalized parent, it says that

$$\mathrm{Sq}(a_0), \quad \mathrm{Sq}(a_1), \quad D_0 \mid T_0, \quad m > 0 \tag{5.11}$$

are jointly solvable if and only if $\mathcal{P}$ has a nonnegative 31-tuple zero. The ratio expressions in (5.3)–(5.6) are precisely those in this interface. This section does not apply a signed interpretation to an unspecified nine-variable polynomial.

The following strengthened completeness fact is needed to replace $b = y_* + 1 + \varepsilon$ by $b = \kappa t^2$.

Lemma 5.2 (Cofinal powers of two)**.** *Given a nonnegative parent witness $\mathcal{P}(1, n_1, \dots, n_\nu) = 0$, the Jones ratio conditions (5.11) have witnesses with $b = 2^a$ for every sufficiently large integer a. In particular, they have witnesses with $b = \kappa t^2$ and $t > 0$.*

Proof. Let $\xi = (1, n_1, \dots, n_\nu)$. The constructive master-system proof `Master3.USys_of_mem` chooses

$$a_0' = y_* + \sum_{r=0}^{\nu} \xi_r + 1, \qquad b = 2^{a_0'}.$$

Inspection of that complete proof shows that the equality specifying a_0' is used only to obtain

$$a \geq 1, \qquad y_* < b, \qquad \xi_r < b \quad (0 \leq r \leq \nu). \tag{C}$$

These conclusions hold for every $a \geq a_0'$, because $a < 2^a$. The rest of the construction depends on (C), on $b = 2^a$, and on the given polynomial zero; it imposes no upper bound on a.

More explicitly, at this chosen base one takes the polynomial evaluations

$$e = e_*(\mathcal{B}), \qquad \ell = \sum_{r=1}^{\nu} \mathcal{B}^{5^r}, \qquad g = \sum_{r=1}^{\nu} \xi_r \mathcal{B}^{5^r}, \qquad \mathcal{B} = b^5,$$

together with q, λ, θ from (5.3). Normalization of the parent implies $g > 0$. The digit bounds and the source's carry-free coding identity give

$$e < 2z_* \mathcal{B}^L, \qquad \ell < 2\mathcal{B}^{5^\nu}, \qquad g < b\mathcal{B}^{5^\nu}.$$

The necessary positive slack is

$$\alpha_{\mathrm{master}} = (b - y_*)q^2 - e\ell g^2.$$

It is positive uniformly in the larger base: putting $r = 5^\nu$,

$$e\ell g^2 < 4z_* b^2 \mathcal{B}^{L+3r} \leq \mathcal{B}^{L+3r+1} \leq \mathcal{B}^{2L} = q^2.$$

Here $4z_* b^2 \leq b^5 = \mathcal{B}$ follows from $b > y_* \geq 2z_*$, and $L + 3r + 1 \leq 2L$ follows from $L = 5r$. The two master quotients

$$\frac{e - y_*}{\theta}, \qquad \frac{\ell - \upsilon_*}{\theta}$$

are positive integers: divisibility is polynomial evaluation modulo $\mathcal{B} - 2z_*$, and positivity follows from $\mathcal{B} > 2z_*$. The three bit-disjointness identities in the master system are the inherited Jones

coding identities; their hypotheses are (C), $b = 2^a$, and the unchanged parent zero. Thus the same explicit construction works at every such larger base.

The proof of `NineReduce.reduced_of_mem` keeps this very base and gives the required packed-rank/binomial condition, as well as $g < b\mathcal{B}^{5^\nu}$. The theorem `NineRatio.ratio_complete` then takes b as a fixed parameter; its hypothesis $32g < G_{\mathrm{cap}}$ follows from that last inequality. It supplies the positive h, s, w, i, j and nonnegative ϕ in (5.11), without selecting a different base.

Write $\kappa = 2^c$, where $c = 4269856(L+1)$. Choose any $r \geq 0$ with $c + 2r \geq a_0'$ and set $t = 2^r$. Then

$$\kappa t^2 = 2^{c+2r}$$

is one of the permitted bases. This gives the required positive t and proves the assertion constructively. □

5.4 Signed coordinates whose guards are unnecessary

Lemma 5.3 (A denominator that never vanishes)**.** *For every integer assignment of the eight base coordinates, $D_0 = JF \neq 0$.*

Proof. The integer $J = 4A - 5$ is nonzero. If $F = (A^2 - 1)E^2 + 1 = 0$, then $(1 - A^2)E^2 = 1$, so $A = 0$ and $E = \pm 1$. But $J = -5$ when $A = 0$, and $E = iJDC^2$ is divisible by 5. This is impossible. □

Lemma 5.4 (The sign of i)**.** *If $i \neq 0$, replacing i by $|i|$ preserves a_0, a_1, D_0, T_0, m, and makes $i_0 = |i| - 1$ nonnegative.*

Proof. The only unsquared use of i is in $E = iJDC^2$. Among the five expressions under consideration, every dependence on E passes through E^2 in F. Thus they depend only on i^2. □

Lemma 5.5 (The h guard follows from the margin)**.** *Suppose $b > y_*$, the five coordinates g_0, s_0, w_0, ϕ, j_0 are nonnegative, a_0 is an integer square, and $m > 0$. Then $h \geq 1$, hence $h_0 \geq 0$.*

Proof. We have $g, s, w, j \geq 1$, $\phi \geq 0$, $b \geq 2$, $\mathcal{B} = b^5 > 2z_*$, $e \geq 0$, and $G_{\mathrm{cap}} > 0$. The margin inequality first gives $K \neq 0$ and

$$g < G_{\mathrm{cap}}/8 = 4b\mathcal{B}^{5^\nu}, \qquad 4(C - KY)^2 < K^2. \tag{5.12}$$

Indeed $m \leq K^2(G_{\mathrm{cap}} - 8g)$; then discarding the strictly positive term $8gK^2$ gives the second inequality.

Put $r = 5^\nu$. Since $\mathcal{B} > 5$ and $b \leq \mathcal{B}$,

$$1 + \mathcal{B} + g < 1 + \mathcal{B} + 4\mathcal{B}^{r+1} \leq 5\mathcal{B}^{r+1} < \mathcal{B}^{r+2} \leq \mathcal{B}^{5r} = q.$$

Also $\ell < \mathcal{B}^{r+1}$, hence $b\ell < \mathcal{B}^{r+2} \leq q < q^3$. The potentially negative part of the first bracket in (5.4) is absorbed by its positive term:

$$\lambda\mathcal{B}q^4 > 2z_*\lambda(1 + \mathcal{B} + g)^4.$$

All the remaining terms of that bracket are nonnegative, and $g > 0$. The second bracket is positive because $q^3 > b\ell$, $\theta > 0$, and $\mathcal{B} > 2$. It follows that

$$R \geq \mathcal{N}^2 - \mathcal{N} \geq 2.$$

Thus $U, Y \geq 1$, $C = 2R + 1 + bw + \phi > 0$, and

$$P = 2R^2Y^2U \geq 2R^2 > R + 2.$$

The second margin consequence rules out $K \le 0$: $K = 0$ is impossible, and $K < 0$ would give $|C - KY| = C + |K|Y > |K|$. Therefore $K > 0$. If $h \le 0$, the relation $K = R + 1 + h(P-1)$ gives $1 \le K \le R + 1 < P$. For $K = 1$ it would imply $P - 1 \mid R$, contradicting $0 < R < P - 1$. For $2 \le K < P$,

$$(PK-1)^2 < (P^2-1)K^2 + 1 < (PK)^2,$$

since the two differences are $K(2P-K) > 0$ and $K^2 - 1 > 0$. This contradicts squarehood of a_0. Consequently $h \ge 1$. □

5.5 The coprime Pell stack

Lemma 5.6 (Six guards in one square test)**.** *Subject to $z_1 \cdots z_6 \neq 0$, squarehood of C_* in (5.7) implies*

$$g_0, s_0, w_0, \phi, j_0 \ge 0, \qquad m > 0. \tag{5.13}$$

Conversely, if these inequalities hold, the six z_r can be chosen nonzero so that C_* is a square.

Proof. Every $B_r \equiv 1 \pmod 4$, so no B_r vanishes. Also $\gcd(B_r, C_{r-1}) = 1$. Inductively, the six factors B_r are pairwise coprime in absolute value. If their product is a square, unique prime factorization shows that every $|B_r|$ is a square. A negative square is 0 or 3 modulo 4, so $B_r \equiv 1 \pmod 4$ forces every factor to be a positive square. If any $d_r < 0$, the nonzero integers C_{r-1}, z_r would give $B_r = 1 + 4d_r C_{r-1}^2 z_r^2 \le -3$, a contradiction. Finally $d_r \equiv 2 \pmod 3$, so it cannot be zero. Thus $d_r > 0$ for all r. For $r \le 5$ this means $x_r \ge 0$; for $r = 6$ it means $m \ge 1$.

For the converse, each $d_r > 0$ and $d_r \equiv 2 \pmod 3$, so it is not a square. The Pell equation

$$a^2 - d_r v^2 = 1$$

has a nontrivial positive solution. It has a positive solution with v divisible by any prescribed positive integer M: the matrix $\left(\begin{smallmatrix} a & d_r v \\ v & a \end{smallmatrix}\right)$ has determinant 1, and some positive power of its reduction modulo M is the identity in the finite group $\mathrm{SL}_2(\mathbb{Z}/M\mathbb{Z})$. Choose $M = 2|C_{r-1}|$, and put $z_r = v/(2C_{r-1})$. Then z_r is a nonzero integer and $B_r = a^2$. Starting with $C_0 = 1$, induction makes every C_r a positive square. □

A useful strengthening is available before any guard inequality is known. If all $z_r \neq 0$, then $d_r \neq 0$, $C_{r-1} \neq 0$, and hence $B_r \neq 1$. The congruence $B_r \equiv 1 \pmod 4$ also excludes $B_r = -1$. Therefore

$$|C_*| > 1 \quad \text{whenever} \quad z_1 \cdots z_6 \neq 0. \tag{5.14}$$

This preguard estimate is what permits the smaller weights.

5.6 Divisibility and nonvanishing in one relation

Lemma 5.7 (An intersective polynomial)**.** *The polynomial*

$$\Pi(T) = (T^3 - T - 1)(T^2 + 23)$$

has no rational zero, but has an integer zero modulo every positive integer modulus.

Proof. The only possible rational roots of the monic cubic are $1, -1$, and neither is a root. The quadratic has no real root.

For an odd prime $p \neq 23$, the cubic has discriminant -23. If it has a root modulo p, the root is simple and lifts to every power of p. If it has no such root, it is irreducible of degree three. Frobenius cycles its three roots in $\mathbb{F}_{p^3}$, an even permutation, so their Vandermonde product lies in $\mathbb{F}_p$. Its square is -23, which is nonzero. Thus $T^2 + 23$ has a simple root modulo p, again

lifting to every power. At $p = 23$, the cubic has the simple root 3, since $3^3 - 3 - 1 = 23$ and $3(3^2) - 1 = 26 \not\equiv 0 \pmod{23}$.

For $p = 2$, use $-23 \equiv 1 \pmod 8$. If an odd a is a root of $T^2 + 23$ modulo 2^e, $e \geq 3$, one of a and $a + 2^{e-1}$ is a root modulo 2^{e+1}: their squares differ by $2^e a$ modulo 2^{e+1}. This supplies roots for all powers of two. The Chinese remainder theorem combines roots for the prime-power factors of any positive modulus. □

Lemma 5.8 (Homogeneous divisibility merger)**.** *For integers D', T', Z' with $D' \neq 0$,*

$$(D' \mid T' \text{ and } Z' \neq 0) \iff \exists v \in \mathbb{Z} : \; Z' D'^5 \mid \mathcal{R}(D', T', v). \tag{5.15}$$

Moreover $\mathcal{R}(D', T', v) \neq 0$ for every integer v.

Proof. The rational identity

$$\mathcal{R}(D', T', v) = D'^5 \Pi(T'/D' + v)$$

proves nonvanishing by Lemma 5.7. If $T' = D'q$ and $Z' \neq 0$, choose v so that $Z' \mid \Pi(q + v)$; this proves the forward implication.

Conversely, $Z' = 0$ would make the displayed divisibility assert $\mathcal{R} = 0$, which is impossible. If $D' \nmid T'$, there is a prime p with $\alpha = v_p(|D'|) > \beta = v_p(|T'|)$. Here $T' \neq 0$. For $r = T' + D'v$, one has $v_p(r) = \beta$. In $r^3 - D'^2 r - D'^3$, the first term has the unique least valuation 3β; in $r^2 + 23D'^2$, it has the unique least valuation 2β. Hence

$$v_p(\mathcal{R}) = 5\beta < 5\alpha = v_p(D'^5),$$

contradicting the asserted divisibility. Signs of D' and Z' do not affect this argument. □

5.7 The three-root norm with signed weights

Lemma 5.9 (The finalizer)**.** *Fix all coordinates except u. For the expressions above, $\exists u \in \mathbb{Z}$: $F_{16} = 0$ is equivalent to*

$$S \neq 0, \qquad S \mid T, \qquad \mathrm{Sq}(C_*), \quad \mathrm{Sq}(a_0), \quad \mathrm{Sq}(a_1). \tag{5.16}$$

Proof. Lemma 5.3 gives $D_0 \neq 0$, and Lemma 5.8 gives $T \neq 0$. If $S = 0$, then $A_0 = A_1 = A_2 = 0$, $X = -T$, and $F_{16} = T^8 \neq 0$. Thus every zero has $S \neq 0$. Since $S = i(z_1 \cdots z_6) D_0^5$, every z_r and i is nonzero; in particular (5.14) applies.

For all real P, K,

$$\sqrt{|(P^2 - 1)K^2 + 1|} \leq P^2 + K^2 + 1. \tag{5.17}$$

Indeed $|(P^2 - 1)K^2 + 1| \leq (P^2 + 1)K^2 + 1$, and the difference between $(P^2 + K^2 + 1)^2$ and this last upper bound is

$$P^4 + K^4 + P^2K^2 + 2P^2 + K^2 \geq 0.$$

It follows from $|C_*| > 1$ that the possibly signed weights satisfy

$$\begin{aligned} |w_1| &> \sqrt{|C_*|}, \\ |w_2| &> \sqrt{|C_*|} + |w_1|\sqrt{|a_0|}. \end{aligned} \tag{5.18}$$

These inequalities require neither positive radicands nor recovered guard inequalities.

Choose complex roots $\rho_0^2 = C_*, \rho_1^2 = a_0, \rho_2^2 = a_1$. Algebraic multiplication gives

$$F_{16} = \prod_{\epsilon_0, \epsilon_1, \epsilon_2 \in \{-1, 1\}} \left[X - S(\epsilon_0 \rho_0 + \epsilon_1 w_1 \rho_1 + \epsilon_2 w_2 \rho_2) \right]. \tag{5.19}$$

For example, first multiplying the ρ_0, ρ_1 signs gives $\left((X \mp \sqrt{A_2})^2 - A_0 - A_1\right)^2 - 4A_0A_1$; the two remaining factors are $W \mp 4VX\sqrt{A_2}$, whose product is (5.10).

A zero makes one of the signed sums in (5.19) equal to $X/S = u - T/S$, a rational number. Distinct sign patterns on nonzero roots give distinct sums: at the largest index where signs differ, the absolute value of the weighted root is at least the absolute value of its weight, since a nonzero integer radicand has root magnitude at least 1. Inequalities (5.18) make this strictly larger than the sum of the magnitudes of all preceding weighted roots. The triangle inequality therefore precludes cancellation. This remains valid for imaginary roots of negative radicands and for negative weights. Zero roots may simply be omitted from this comparison.

The field $\mathbb{Q}(\rho_0, \rho_1, \rho_2)$ is the splitting field of three quadratic polynomials, hence a finite Galois extension of $\mathbb{Q}$. Each automorphism changes root signs. Since the encoded sum is rational, every automorphism fixes it. The separation just proved forces every nonzero root in it to be fixed individually. Therefore all three roots are rational. A rational root of a monic polynomial $T^2 - a$, $a \in \mathbb{Z}$, is an integer. Thus C_*, a_0, a_1 are nonnegative integer squares. The encoded sum is now an integer, so T/S is an integer as well. This proves necessity of (5.16).

Conversely, under that interface choose integer square roots and set

$$u = T/S + \rho_0 + w_1\rho_1 + w_2\rho_2.$$

This is an integer and makes a factor in (5.19) vanish. □

5.8 Soundness, completeness, and the consistency equivalence

Theorem 5.10 (The sixteen-integer presentation)**.** *For the fixed coefficients above, ordinary mathematics formalizable in ZFC proves*

$$(\exists \mathbf{x} \in \mathbb{Z}^{16} : F_{16}(\mathbf{x}) = 0) \iff (\textit{there is a finite syntactic ZFC refutation}).$$

Consequently,

$$\mathrm{ZFC} \vdash (\nexists \mathbf{x} \in \mathbb{Z}^{16} : F_{16}(\mathbf{x}) = 0) \iff \mathrm{Con(ZFC)}.$$

Proof. For soundness, let $F_{16} = 0$. Lemma 5.9 gives $S \neq 0$, $S \mid T$, and the three square conditions. Lemma 5.8 gives $D_0 \mid T_0$ and $iz_1 \cdots z_6 \neq 0$. Lemma 5.6 recovers the five nonnegativity guards and $m > 0$.

At this point $t \neq 0$ follows without a separate nonzero factor. If $t = 0$, then $b = \kappa t^2 = 0$ and $G_{\mathrm{cap}} = 0$, while $g = g_0 + 1 \geq 1$. The literal margin formula would give

$$m = -8gK^2 \leq 0,$$

a contradiction. Hence $b = \kappa t^2 \geq \kappa > y_*$, and $\varepsilon = b - y_* - 1$ is a nonnegative integer. Lemma 5.5 gives $h_0 \geq 0$. Lemma 5.4 replaces i by $|i|$, preserving all four Jones conditions and making $i_0 \geq 0$. We have therefore recovered an admissible nonnegative tuple for the unchanged Jones interface (5.11). The inherited ratio soundness theorem gives a zero of the parent quartic. Its complete native/Pell compiler interface gives a blank K215 halting computation, and the calibrated K215 theorem gives a finite syntactic ZFC refutation.

For completeness, a finite ZFC refutation gives a blank K215 halting run and then a positive parent-quartic solution by those same two interfaces. After the coordinate shift to the normalized parent, apply Lemma 5.2. It gives a Jones tuple satisfying (5.11) at $b = \kappa t^2$ with $t > 0$, all the ordinary base coordinates nonnegative, and $i > 0$. Lemma 5.6 supplies nonzero $z_1, \ldots, z_6$ for which C_* is a square. Now $Z = iz_1 \cdots z_6 \neq 0$; the inherited Jones tuple has $D_0 \mid T_0$. Lemma 5.8 chooses v with $S \mid T$. The three radicands are squares, so Lemma 5.9 chooses the final integer u and gives $F_{16} = 0$.

Every extra root, quotient, or exponent selected in this proof is a proof witness used to establish equivalence. The displayed equation itself quantifies only the sixteen coordinates in (5.2). Negating the existence equivalence gives the stated consistency equivalence. Neither direction assumes or proves Con(ZFC). □

5.9 Degree and arithmetic size

All degrees in this subsection give each of the sixteen unknowns degree one and every fixed coefficient degree zero. The largest term in the packed rank is $\lambda\mathcal{B}q^{11}\mathcal{N}^2$; it gives $\deg R = 470L$. The largest term in the margin is $-4K^2Y^2G_{\mathrm{cap}}$, giving $\deg m = 4442L + 12$. The useful intermediate degrees are

R	$470L$	P	$1900L + 3$
K	$1900L + 4$	A	$1110L + 2$
a_0	$7600L + 14$	a_1	$49280L + 68$
m	$4442L + 12$	C_5	363
C_*	$4442L + 1103$	D_0	$13750L + 20$
S	$68750L + 107$	T	$68750L + 105$
X	$68750L + 108$	w_2	$8242L + 1111$
A_0	$141942L + 1317$	A_1	$153984L + 2434$
A_2	$203264L + 2504$		

For the first five stack stages the degree recurrence is $c_r = 3c_{r-1} + 3$, starting at $c_0 = 0$. The last stage has degree $3 \cdot 363 + (4442L + 12) + 2 = 4442L + 1103$. The seven factors of $Z = i\prod_{r=1}^{6} z_r$ account for the constant $107 = 5 \cdot 20 + 7$ in the degree of S.

The term A_2 has strictly greater degree than X^2, A_0, A_1. Thus V has the leading form of A_2, W that of A_2^2, and the final polynomial that of A_2^4. All of these leading forms are nonzero: the rank has the displayed unique top term, the margin has its unique negative top term, and subsequent relevant leading forms are products and squares of nonzero forms. In particular, after the authentic positive fixed coefficients are substituted,

$$\deg F_{16} = 4(203264L + 2504) = 813056L + 10016.$$

This is an exact degree assertion, not just a structural upper bound that ignores cancellations.

The literal pruned evaluator contains

$$697 \text{ additions } + \ 26 \text{ subtractions } + \ 894 \text{ multiplications } = \ 1617 \text{ operations.}$$

Each binary operation, including a square and multiplication by a fixed numeral, is charged. Reuse and fixed leaves are free, consistently with the repository's arithmetic-circuit convention. Fixed exponentiation is expanded into multiplication gates; no exponentiation oracle is used.

For clarity, the compact evaluation of the enormous digit baseline does not omit its zero-coefficient positions. If $p_r = \mathcal{B}^{5^r}$, then the baseline is represented through the complete homogeneous polynomial of degree four, and

$$\lambda = \left(\prod_{r=0}^{\nu}(1 + p_r + p_r^2 + p_r^3 + p_r^4)\right)(1 + q)(1 + q^2).$$

These are polynomial identities. The full baseline contains $\binom{36}{4} = 58905$ distinct positions; the 402 nonzero weighted-coefficient records supply its sparse correction.

5.10 Verification boundary

The final binding is `signed16/data/INSTANCE.json`; it names the whole compiler tuple, parent, weighted-coefficient records, and index recipes. The literal division-free polynomial is `signed16/generated/signed16.json`. The new reduction does not claim a fresh end-to-end Lean or Rocq certificate. The source Lean files explain the precise Jones interfaces and the cofinal construction, while the native compiler and its K215 calibration remain identified mathematical proof dependencies.

The independent data-only auditor imports neither the emitter nor its DAG utility. It checks all four decompressed port hashes, enumerates the 58905 baseline positions directly, evaluates the geometric sum by a separate method, and compares 288 named-node values on six signed assignments over two primes, including $t = 0$. It also checks the eight-factor norm in the rank-eight ring with three formal square roots on 100 exact assignments. For the final numerical instance, the top-degree diagonal coefficients are nonzero modulo two primes:

$$121738 \pmod{1000003}, \qquad 73024 \pmod{1000033}.$$

Together with the propagated degree upper bound, either nonzero residue is an independent certificate that the top degree survives on this fixed coefficient slice.

These checks establish transcription, coefficient binding, and algebraic metrics. The universal soundness and completeness claims rest on the proofs above and the stated inherited interfaces, not on finite tests or a search for a Diophantine solution.

6 Smaller positive systems and circuit evaluations

The coefficient adapter also supports two improvements that preserve the positive-domain arithmetic constructions of package 30 [7]. One retains degree four and reduces the arithmetic evaluation cost; the other retains twelve unknowns and reduces the cost of its much higher-degree norm. All the identities in this section hold with the four coefficient ports treated as independent indeterminates. We then specialize *all four* ports to the new complete 121 by 8801 tuple. The resulting fixed polynomials need not equal the historical fixed polynomials, because their coefficients have changed. Their arithmetic templates have the exact identities proved below.

Presentation	Unknowns	Equations	Degree	Operations
Quadratic comparison system	31	22	2	60
Its sum of squares	31	1	4	125
Twelve-variable product	12	1	1081	108
Twelve-variable system	12	3	1072	103
Transport-shifted system	12	3	1072	102

Every unknown in this table is a strictly positive integer. For a system, “degree” means the maximum degree of a residual. Every binary addition, subtraction and multiplication is charged, including a multiplication by a fixed numeral; a comparison and reuse of an existing register are free. The source costs at the same two points were 65/130 operations for the quadratic system/quartic, and 107/111 for the twelve-variable system/product. These are local improvements. The separate inherited 59-operation higher-degree systems and 67-operation single equation retain their lower total-operation counts.

6.1 A 60-operation quadratic system

The 31 supplied positive integers are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta.$$

Use the fixed ports β, K_0, M_C, M_F, where M_F is the already shifted high mask. Introduce only the polynomial abbreviations

$$q = \beta J + 1, \quad k = \eta + \zeta, \quad H = 4a + 3, \quad C = q - F - \alpha, \quad U = L + 1,$$
$$Y = a - E, \qquad Q = c - \eta, \qquad T = P - k, \qquad R = jc - V.$$

In particular, R is quadratic. The following are the 22 comparisons:

$$\begin{aligned}
&U = q^2, && N = qU, && X = wq, && Y = sN,\\
&E = XY, && Q = kY, && T = EQ, && \tau^2 = PT + 1,\\
&D = X + ac + \gamma H, && \Delta = a^2 + H, && C_2 = c^2, &&\\
&D^2 = \Delta C_2 + 1, && k = R + 1 + hE. &&&&
\end{aligned} \tag{6.1}$$

$$\begin{aligned}
R &= (q - F)(N - q) - CL + (M_C + qM_F)J,\\
(K_0 + w)C &= F + z\beta J,\\
F_2 &= f^2, \qquad V = of - c, \qquad g = y^2 - V^2,\\
I &= iC_2, \qquad B_{\mathrm{aux}} = I^2, \qquad B_{\mathrm{aux}} = \Delta(F_2 - 1),\\
y^2 &- B_{\mathrm{aux}} g = 1.
\end{aligned} \tag{6.2}$$

Each equality has a quadratic residual after substituting the displayed abbreviations. If these residuals are $r_1, \ldots, r_{22}$ in the displayed order, the single polynomial is

$$\boxed{Q_{125} = \sum_{j=1}^{22} r_j^2.} \tag{6.3}$$

Thus its positive zero set is exactly the system's positive solution set.

The first four simplifications reorient existing equalities so that a computed difference or sum serves two purposes. The equality $L = q^2 - 1$ becomes $L + 1 = q^2$, and the same $L + 1$ occurs in $N = q(L + 1)$. The equality $a = E + sN$ becomes $a - E = sN$, reusing the $a - E$ in $E = X(a - E)$. Similarly, $c - \eta = k(a - E)$ reuses $c - \eta$ in the equation defining P, and $P - k = E(c - \eta)$ reuses $P - k$ in $\tau^2 = P(P - k) + 1$. These are polynomial identities on all integer assignments.

The fifth simplification is the positive transport-coordinate shift also used in package 26 [4]. Its original equation is

$$(K_0 + w)C + (q - F) - z_{\mathrm{old}}(q - 1) = 1.$$

Substituting $z_{\mathrm{old}} = z + 1$ and $q - 1 = \beta J$ gives exactly the second equation in (6.2). A new positive solution therefore gives an old one. Conversely, first apply the established compiler reconstruction to an old solution. It gives

$$C > 0, \qquad F < q, \qquad X = wq = 2^R, \qquad R > 3q,$$

and hence $w > q > F$. The original transport equation implies

$$(z_{\mathrm{old}} - 1)(q - 1) = (K_0 + w)C - F > 0.$$

Thus $z_{\mathrm{old}} \geq 2$, and $z = z_{\mathrm{old}} - 1$ remains positive. This establishes a bijection of positive zero sets. The positivity claim is a consequence of the completed source reconstruction; it is not presumed in the initial bootstrap.

The resulting comparison circuit has 33 multiplications and 27 additions/subtractions, hence 60 operations. Forming the residuals, squaring them and summing gives 55 multiplications and 70 additions/subtractions, hence 125 operations. Its degree is exactly four: τ occurs only in the residual $\tau^2 - P(P-k) - 1$, so the coefficient of τ^4 in (6.3) is exactly one. This argument is uniform in every fixed coefficient slice and does not depend on a generic numerical specialization.

The supplied builder checks exact equality of all 22 residuals with the source residuals after $z_{\mathrm{old}} = z + 1$, treating all four ports symbolically. It expands the quartic to 299 terms when port exponents are included in the symbolic monomials. Specializing actual coefficients can identify terms, so 299 is stated with that convention. The degree-four certificate is unaffected.

6.2 A 108-operation twelve-variable equation

The twelve supplied positive integers are now

$$J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t.$$

The notation in this subsection is local. Set

$$\begin{array}{lllll} q = \beta J + 1, & C_w = q - F - \alpha, & X = wq, & Y = sq^3, & E = XY, \\ k = \eta + \zeta, & c = kY + \eta, & a = E + Y, & H = 4a + 3, & \Delta = a^2 + H, \\ \ell = EkY, & A = 1 + \ell(\ell + k), & G = 1 + \Delta c^2, & b_0 = X + ac. & \end{array}$$

The transport and index registers are

$$\begin{aligned} R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\ N_k &= k - hE - R, \qquad N_t = (K_0 + w)C_w + q - F - z(q-1). \end{aligned}$$

For the Pell norm define

$$\begin{gathered} I = ic^2, \quad D = 1 + \Delta I^2, \quad Q = \Delta(D-1), \quad d = Q - 1, \quad b = c + RD, \\ M = k(I^2 + T^2), \qquad L = HM + 1, \qquad W = b_0 + Ht. \end{gathered}$$

The economical scalar evaluator is

$$\begin{aligned} S &= H^2 Q, \qquad H_P = SDc^2T^2, \qquad H_U = Sb^2, \\ n &= (HM)^2 A, \qquad d_n = dn, \qquad H_e = H_P + H_U - H^2, \\ j &= 4H_P H_U, \qquad \mathcal{T} = 2d(W^2 + L^2 G), \\ r &= d(W^2 - L^2 G), \qquad \mathcal{K} = r^2, \\ a_1 &= \mathcal{T} - 2(d_n + H_e), \qquad b_1 = (d_n - H_e)^2 - j - \mathcal{K}, \\ v &= a_1(a_1\mathcal{T} + 2b_1) - 16 d_n j, \\ u &= a_1^2 \mathcal{K} - b_1^2. \end{aligned}$$

All these letters denote polynomial abbreviations, not new witnesses. Finally put

$$N = \mathcal{K}v^2 - u(\mathcal{T}v - u), \qquad \boxed{P_{108} = N_k N_t (1 + N) - 1.} \tag{6.4}$$

Its equation is $P_{108} = 0$. Its three-equation version is

$$N_k = 1, \qquad N_t = 1, \qquad \mathcal{K}v^2 = u(\mathcal{T}v - u). \tag{6.5}$$

Replacing the middle equation by

$$(K_0 + w)C_w = F + z(q - 1) \tag{6.6}$$

gives the 102-operation system. Its z is the old transport coordinate minus one; the positivity and equivalence proof is exactly the one above, applied after the twelve-variable reconstruction.

6.3 Why the compressed norm is exact

The source scalar norm is the product over the main square-root conjugates of a polynomial encoding the first, strong and auxiliary Pell conditions. For its quadratic-remainder evaluator, the two final coefficients are

$$v_r = a_1(a_1\mathcal{T} + 2b_1) - 16d_n j, \qquad u_r = b_1^2 - a_1^2\mathcal{K},$$

and the final norm is

$$u_r^2 + \mathcal{T}u_r v_r + \mathcal{K}v_r^2.$$

The identity

$$u_r^2 + \mathcal{T}u_r v_r + \mathcal{K}v_r^2 = \mathcal{K}v^2 - u(\mathcal{T}v - u), \qquad u = -u_r, \quad v = v_r, \tag{6.7}$$

holds in every commutative ring. The sign change reverses an existing subtraction, and the factorization saves one multiplication.

Two additional ring identities give the other savings. Sharing H^2Q between $H_P = H^2QDc^2T^2$ and $H_U = H^2Qb^2$ saves one multiplication. Computing

$$d^2(W^2 - L^2G)^2 = [d(W^2 - L^2G)]^2$$

saves another because the source's separate d^2 has no other consumer. Thus the product takes $108 = 65 + 43$ operations, the summands being multiplications and additions/subtractions. The final comparison in (6.5) avoids one more subtraction, giving $103 = 63 + 40$. The transport shift gives $102 = 63 + 39$.

No new rational-root, root-separation or Pell-saturation hypothesis is introduced by these changes. The norm is the identical four-port polynomial template. Consequently the source's universal reconstruction, its product-unit sign argument and its nonzero leading homogeneous form remain valid. The norm has exact degree 1072, and N_k, N_t have degrees $7, 2$, respectively, so (6.4) has exact degree 1081. The active coefficient tuple has $\beta > 0$, which is the coefficient condition needed for the inherited noncancellation certificate.

6.4 Logical consequence and reproducibility

For each presentation in this section, the new complete coefficient adapter, the inherited arithmetic reconstruction, and the literal K215 calibration give

$$\begin{aligned} \text{positive solvability} &\iff \text{blank halting of K215} \\ &\iff \text{existence of a finite ZFC refutation.} \end{aligned}$$

Negation yields the requested equivalence with Con(ZFC) inside ZFC. The construction establishes neither side of that equivalence.

The quartic generator performs exact sparse-polynomial verification of all residuals and of the degree certificate. The twelve-variable checker verifies five local ring identities exactly, audits every live input and operation, compares 240 signed modular assignments with the source template, and checks twelve additional assignments using residues of the actual bound coefficients.

Both active bindings authenticate the same four payloads. Their historical source files have been restored to exact repository bytes and checked against the pinned Git blob identifiers. The new algebraic and coefficient arguments remain ordinary mathematical proofs; these finite checks are not a new end-to-end Lean or Rocq theorem.

7 Reproduction, attribution and remaining work

7.1 What is in the archive

The LaTeX source consists of `report.tex`, the files under `sections/`, and `signed10/SIGNED10_SECTION.tex`. The compiled `report.pdf` contains the complete mathematical specification. The four implementation directories serve distinct purposes:

`signed10/` The ten-integer equation, four comparison variants, explicit fixed-leaf recipes, norm polynomials, coefficient payloads, a builder, and author and independent audits.

`signed16/` The independent Jones-based sixteen-integer equation, its index data, coefficient recipes, circuit and verifier.

`positive12/` The 108-operation product, 103-operation unshifted system and 102-operation shifted system, with symbolic and concrete-instance checks. It also contains an independent audit of the new native coefficient construction.

`quartic31/` The 60-operation quadratic system, 125-operation quartic and exact symbolic coefficient data, together with an independent literal audit of the sixteen-variable equation.

`coefficients/` The literal machine transitions and tile table, the complete finite tuple recipe, raw and compressed port bytes, modular/controller certificates, and the adapter proof.

Selected pinned source proofs are under `sources/` and the local `upstream/` directories. They are evidence and context; they are not a standalone checkout of ProveIt or all of its Lean dependencies. The original machine notices and repository license are preserved.

7.2 Commands and verification scope

From the extracted archive root, run

```
python3 verify_all.py
make pdf
```

The arithmetic checks require Python 3.11 or later and only its standard library. They need no network access or solver. Do not use Python's `-O` option: many exact assertions are part of the audit, and the top-level driver rejects optimized execution. The PDF build requires a standard pdfLaTeX installation with the packages listed in the preamble. The scripts resolve their data relative to their own locations, so the tree can be moved into a new numbered ProveIt report directory unchanged.

The finite audits establish the following concrete facts. Every emitted operation has two earlier inputs and is an addition, subtraction or multiplication. Every advertised variable is live. All actual port bytes and their hashes agree with the complete tuple. The parent normalization reconstructs the 402 weighted coefficient records exactly. The index and Sun-source bounds hold for those actual integers. The three-root and four-root norm identities are checked as exact polynomial identities. Separate literal evaluators agree with the emitted circuits on signed

and boundary assignments modulo several primes. These assignment checks supplement the mathematical transcription proofs; they do not replace them.

Degree verification is stronger than a collection of sample values. Gate-by-gate propagation gives an upper bound. Substituting a common indeterminate into the variable leaves, while retaining exact leading coefficients modulo a specified prime, produces a nonzero coefficient at the claimed top degree. Thus the bound is attained for the actual fixed tuple. The low-degree systems additionally have exact symbolic residual comparisons; the quartic has the uniform coefficient $[\tau^4]Q = 1$.

7.3 What each improvement contributes

The native tuple combines the unchanged mixed-radix tile geometry with a different odd-factor rank controller. Its contribution is smaller fixed ports, not a smaller machine or a different simulated language. The ten-integer equation specializes an existing universal reduction to this literal source, then reduces the number of square tests, the homogeneous merger degree and the root-separation weights. The sixteen-integer equation develops a separate base restriction and guard elimination; its large circuit improvement also uses the already established digit evaluation identities from package 27. The 12- and 31-variable changes are exact algebraic refactorings and a justified positive-coordinate shift. Their savings should be interpreted at their stated degree and domain, not as a new universal minimum operation count.

7.4 Useful next steps

The most valuable next verification step is to formalize the new coefficient adapter, the signed norm lemma, the quintic merger and the cofinal-base lemma, then connect their literal circuit definitions to the existing K215 endpoint. For the ten-variable route the modified Lucas-certificate and residue-interpolation theorems of the 2026 preprint also belong in that formalization boundary.

Further arithmetic sharing may reduce the 1,442-operation evaluator; no addition-chain or circuit minimality certificate is supplied. A smaller native machine, a different tile language, or a smaller suitable low-degree source could improve several metrics at once, but would require a new calibrated compiler argument. Reducing ten unrestricted integer variables is a separate research problem. The present work does not claim that the general ten-unknowns bound, the nine-variable natural-number bound, or this particular consistency instance is optimal.

References

[1] V. Reshetnikov and contributors, *ProveIt: Con(ZFC) Diophantine forms*, source snapshot `8c6eec9d19ef80538c8e61ff6e0180040583446a`. https://github.com/VladimirReshetnikov/ProveIt/tree/8c6eec9d19ef80538c8e61ff6e0180040583446a/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/con-zfc-diophantine-forms.

[2] ProveIt, package 24, *integer24*; in particular the general odd-factor controller in its coefficient construction, at the snapshot in [1]. Relevant source is included under `coefficients/upstream/24-coefficients.tex`.

[3] ProveIt, package 25, *hybrid12*, mixed-radix Wang943 coefficient construction, at the snapshot in [1]. Included source: `coefficients/upstream/25-COEFFICIENT_PROOF.md`.

[4] ProveIt, package 26, *ratio12*, including the signed product and ratio normalizations, at the snapshot in [1].

[5] ProveIt, package 27, *wave4*, digit-evaluation circuits and coefficient/cost comparisons, at the snapshot in [1].

[6] ProveIt, package 29, *simpler-forms*, including the 17-integer Jones wrapper, at the snapshot in [1]. Included source: `sources/SIGNED17_SOURCE_PROOF.md`.

[7] ProveIt, package 30, *quartic31*: the positive quartic parent, the normalized 31-variable source, and the 12-variable specialization, at the snapshot in [1].

[8] ProveIt, `KEndpoint215.lean`, calibrated blank-input K215 endpoint, at the snapshot in [1]. The exact source used here is bundled as `sources/KEndpoint215.lean`.

[9] J. P. Jones, Universal Diophantine equation, *Journal of Symbolic Logic* **47** (1982), 549–571. https://doi.org/10.2307/2273588.

[10] Z.-W. Sun, Further results on Hilbert's Tenth Problem, *Science China Mathematics* **64** (2021), 281–306. Version used: arXiv:1704.03504v7, especially the explicit construction in Section 3 and the square-combining polynomial in Section 5. https://arxiv.org/html/1704.03504v7.

[11] Z.-W. Sun, Ten unknowns for Hilbert's tenth problem over the integers, preprint, arXiv:2609.01594v2 (2026), especially Sections 3–7. https://arxiv.org/html/2609.01594v2.